A SIX-POINT POSITIVE-WEIGHT OBSTRUCTION TO DEGREE-FOUR HERMITE CURVATURE REPAIRS ON MANIFOLDS WITHOUT CONJUGATE POINTS

ARGUS OPEN RESEARCH PROJECT

Abstract. In the construction of Riemannian metrics without conjugate points whose geodesic rays fail opposite-end divergence, we prove that the degree-four compactly supported Hermite transverse blister repair over $[-16, 16]^5$ cannot satisfy the preregistered sectional-curvature gate $K \le 1$: six positive rational plateau weights simultaneously annihilate all five free coefficient directions and yield a strictly negative weighted slack, forcing every real coefficient vector in $\mathbb{R}^5$ to violate $K \le 1$ at one of six exact rational evaluation points.

1. Main Farkas Obstruction

Theorem 1.1. *Let g_c ($c \in \mathbb{R}^5$) be the five-parameter degree-four Hermite compactly supported metric perturbation preserving the full first jet along the central geodesic. There exist six explicit rational evaluation points $q_1, \dots, q_6$ and positive rational weights $w_1, \dots, w_6 \in \mathbb{Q}_{>0}$ such that the linearized coefficient matrix $B \in \mathbb{Q}^{6\times 5}$ satisfies $w^{\mathsf{T}} B = 0$ and the baseline curvature excess vector b_0 satisfies $w^{\mathsf{T}} b_0 > 0$. Consequently, for every $c \in \mathbb{R}^5$, $\max_{1\le i\le 6} K_{g_c}(q_i) > 1$.*

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 20001904 in the domain of Riemannian geometry follows a strict separation between the exact statement decided and nearby formulations with stronger hypotheses. Specifically:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, or symbolic arithmetic without floating-point approximation.

(c) *Classification:* Within the program taxonomy, this result is categorized as a *scoped farkas obstruction* deciding the scope: six positive rational plateau weights annihilate all 5 Hermite coefficient directions and force K > 1 on [-16,16]^{ }5.

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that are *not* decided by the argument above:

- unrestricted geodesic ray divergence on manifolds without conjugate points.

References

[1] Primary problem list and source literature for Result 20001904 (Riemannian geometry), archived with exact theorem locators and verification certificates in the accompanying repository package.

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